i don't understand if papers use abramson as a test benchmark, and after implementing their methodologies, they could be able to reproduce abramson experiments

Clarifying how later papers use the Abramson hydrofoil case

The key distinction is between:

- **Using Abramson's data to tune/validate a method**, and
- **Designing a method that, in principle, could reproduce such an experiment.**

1. Direct use as a benchmark case

Chu & Abramson (1969) explicitly take the *same fully submerged subcavitating hydrofoil model* and recompute its flutter speed. They vary **lift-curve slope** and **center-of-pressure location**, and when these are combined with a **relaxation of the Kutta condition**, they obtain "excellent agreement with the measured flutter speed" of Abramson's experiment (Chu & Abramson, 1969; Abramson & Ransleben, 1964).

- This paper clearly treats Abramson's case as a **benchmark** and shows that, after modifying the unsteady aerodynamics, the method can match the original result.

Low mass ratio hydrofoil flutter model design chooses the **Abramson & Ransleben experimental campaign as the main reference** for model geometry, mass ratio (~0.99), setup, and test procedures (D'Ubaldo et al., 2022).

- They state that Abramson & Ransleben's report is complete and accurate and use it to guide design and to keep their model "as close as possible" in profile and mass ratio (D'Ubaldo et al., 2022).
- This is **benchmarking at the experimental-design level**, not a numerical reproduction of the old data.

2. Methods that could (in principle) reproduce similar cases

Several works develop high-fidelity **fluid–structure interaction** or **viscous unsteady aerodynamic** models aimed at accurately predicting flutter/divergence of flexible foils in dense or viscous fluids (Chae et al., 2013; Akcabay & Young, 2014; Taha & Rezaei, 2019; Taha & Rezaei, 2020; Santos et al., 2020).

- These models are intended precisely to fix the type of prediction failures seen in 1960s flutter tests (Taha & Rezaei, 2019).
- However, in the provided texts there is **no statement** that they actually re-run or quantitatively validate on the Abramson hydrofoil case.

3. What can be safely concluded

Question	Evidence-based answer	Citations
Is Abramson used as a benchmark?	Yes: explicitly by Chu & Abramson (1969) and as the main experimental reference in the 2022 low-mass-ratio model design.	(Chu & Abramson, 1969; D'Ubaldo et al., 2022; Abramson & Ransleben, 1964)

Question	Evidence-based answer	Citations
Do newer general methods *demonstrate* reproduction of Abramson data?	They are motivated by similar failures of classical flutter theory, but the texts shown do not claim validation on the Abramson case.	(Chae et al., 2013; Akcabay & Young, 2014; Taha & Rezaei, 2019; Taha & Rezaei, 2020; Santos et al., 2020)

FIGURE 1 How later works reference or extend Abramson's flutter case

Summary

Only Chu & Abramson (1969) clearly use the original hydrofoil as a **numerical benchmark** and show that their modified theory reproduces the measured flutter speed. The 2022 experimental design paper uses Abramson & Ransleben as a **primary experimental template**. Other modern FSI and viscous-aerodynamics papers present methods that could, in principle, handle such a case, but the available abstracts do not show them explicitly benchmarking on Abramson's experiment.

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